MIL-E-1/441B
6 June 1975
SUPERSEDING
MIL-E-1/441A
31 January 1968

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 35W4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, miniature, high-vacuum, half-wave

Outline --- 5-3 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	nc	nc	h	h	a	ht	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	Epp	ib	Io	Ri	Ci	i(surge)	Ehk	Alt
Unit:	V	v	Vac	ma	mAdc	Ohms	µF	a	v	ft
Maximum:	38.5	365	---	660	110	---	---	2.2	330	Note 1
Minimum:	31.5	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	35	---	130	---	---	1,050	40	---	---	---

(B) (without panel lamp)

GENERAL:

Qualification - Required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1231	Emission	-	Eb = 30 Vdc	0.65	II	Is	250	---	mAdc
1353	Operation of rectifiers	2,3		0.65	II	Io	105	---	mAdc
---	Heater voltage tap	4		0.65	II	---	6.5	8.5	V
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Heater current	-		---	---	If	138	162	mA
1336	Heater-cathode leakage	-	Ehk = -300 Vdc	---	---	Ihk	---	200	µAdc
2126	Glass strain	-		4.0	I	---	---	---	---
1121	Base strain	-		---	---	---	---	---	---
1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Life test (intermittent)	5	Group B	---	---	---	---	---	---
---	Life-test end point: (intermittent) (500 hours)								
1353	Operation of rectifiers	-		---	---	Io	90	---	mAdc

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. In a half-wave circuit, adjust Zp such that a tube having Etd = 18 Vdc at 200 mAdc gives Io = 110 mAdc. Ehk = Eo + 117 Vac.
4. Voltage is measured between pins 4 and 6. Heater voltage tap test may be made a quality conformance inspection, part 2 test, with an AQL of 6.5, inspection level S3, provided a short and discontinuity detection test (method 1201) is performed on the total production.
5. In life-test operation, the values of Rt and Ct given in test conditions may be considered as approximate, and shall be adjusted initially to give Io = 110 mAdc with ib greater than 600 ma. Ehk = Eo + 117 Vac.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES
Army - AV

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2846)

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1111B ELECTRON TUBE, RECEIVING TYPE 35WL

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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